AI-Based PDF & Text Summarization Platform

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1. Introduction

The AI-Based PDF & Text Summarization Platform is a full-stack web application developed to simplify document understanding and text summarization using Artificial Intelligence.

The application allows users to:

- Upload PDF documents
- Extract text from uploaded PDFs
- Generate AI-powered summaries
- Summarize custom text input
- Chat with uploaded PDF documents
- View multi-page PDFs directly in the browser
- Download AI-generated summaries as professionally formatted PDF reports

The project combines Natural Language Processing (NLP), Large Language Models (LLMs), and full-stack web development to create an intelligent document assistant.

This platform is useful for students, researchers, professionals, and anyone who wants to quickly understand lengthy documents.

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2. Objectives

The primary objectives of this project are:

- To build an AI-powered text summarization platform
- To enable PDF upload and text extraction
- To generate concise summaries using AI models
- To provide an interactive chat system for uploaded PDFs
- To create a clean and user-friendly interface
- To demonstrate full-stack AI application development
- To allow users to download AI-generated summaries

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3. Problem Statement

Reading and understanding large PDF documents is time-consuming and inefficient.

Users often need quick summaries and answers from documents without reading the entire content manually.

The goal of this project is to automate document understanding using AI technologies by:

- Extracting important information
- Generating concise summaries
- Enabling document-based question answering

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4. Technologies Used

Frontend

- React.js
- React Hooks
- react-pdf
- JavaScript
- CSS Inline Styling

Backend

- Node.js
- Express.js
- REST APIs

AI & NLP

- Groq API
- Llama 3.1 AI Model

PDF Processing

- pdf-parse
- multer

Development Tools

- Visual Studio Code
- Git & GitHub
- Postman

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5. System Architecture

The system consists of two major components:

Frontend (React.js)

Responsible for:

- User interface
- PDF upload
- Displaying summaries
- Displaying chat messages
- Rendering PDF documents

Backend (Node.js + Express)

Responsible for:

- Handling API requests
- Processing uploaded PDFs
- Extracting PDF text
- Connecting with AI APIs
- Generating summaries and chat responses

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6. Project Features

6.1 PDF Upload

Users can upload PDF documents directly from the browser.

The uploaded file is stored temporarily on the server.

6.2 PDF Text Extraction

The system extracts text from uploaded PDF files using the pdf-parse library.

The extracted text is split into chunks for efficient processing.

6.3 AI PDF Summarization

After uploading a PDF, the extracted content is sent to the Groq AI API.

The AI model generates a concise summary of the uploaded document.

6.4 AI Text Summarization

Users can manually enter text into a textarea.

The AI model summarizes the entered text instantly.

6.5 Chat With PDF

Users can ask questions related to the uploaded PDF.

The AI responds using the extracted document content.

6.6 Multi-Page PDF Viewer

The project supports viewing multiple PDF pages directly inside the application using react-pdf.

6.7 Download Summary as PDF

Users can export/download AI-generated summaries as professionally formatted PDF reports.

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7. Workflow

Step 1

User uploads a PDF document.

Step 2

Backend extracts text from the PDF.

Step 3

Extracted text is indexed and stored temporarily.

Step 4

Groq AI API generates a summary.

Step 5

Summary is displayed on the frontend.

Step 6

User can ask questions related to the uploaded document.

Step 7

AI generates answers based on the uploaded PDF content.

Step 8

Users can download AI-generated summaries as professionally formatted PDF reports.

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8. API Endpoints

8.1 Upload PDF

POST /api/upload

Uploads PDF and generates summary.

8.2 Text Summarization

POST /api/summarize

Generates AI summary from custom text.

8.3 Chat With PDF

POST /api/chat

Answers questions related to uploaded PDF.

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9. Folder Structure

```
AI-Based-Text-Summarization-Platform/
|
 ├── frontend/
|   ├── public/
|   ├── src/
|   |   ├── components/
|   |   |   ├── Chat.js
|   |   |   └── PDFViewer.js
|   |   ├── App.js
|   |   ├── index.js
|   |   └── index.css
|
| ├── backend/
|   ├── routes/
|   |   ├── upload.js
|   |   ├── summarize.js
|   |   └── chat.js
|   ├── services/
|   |   └── vectorStore.js
|   ├── uploads/
```

```
|   ├── texts/
|   ├── server.js
|  └──. Env
|
├── Screenshots/
|   ├── UI.png
|   ├── file_uploaded_UI.png
|  └── chatwithpdf.png
├── README.md
└── Project Report.pdf
```

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10. Challenges Faced

During development, several technical and integration challenges were encountered:

• Handling large PDF files efficiently

• Managing AI API response formatting and cleanup

• PDF parsing inconsistencies in multi-page documents

• Multi-page PDF rendering using react-pdf

• Managing previous PDF data in memory during new uploads

• Frontend and backend API integration

• Preventing markdown formatting issues in AI-generated responses

• Formatting exported PDF summaries professionally

• Handling asynchronous API requests and loading states

These challenges were resolved through debugging, backend optimization, improved state management, AI response sanitization, and frontend rendering improvements.

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11. Live Deployment

The project has been successfully deployed online and is accessible through the following live application link:

Live Application:

https://ai-based-text-summarization-platfor.vercel.app/

The deployed platform allows users to upload PDF documents, extract text from uploaded PDFs, generate AI-powered summaries, summarize custom text, interact with uploaded PDF documents through chat, view multi-page PDFs directly in the browser, and download AI-generated summaries as professionally formatted PDF reports.

For deployment, the frontend application is hosted on Vercel, while backend APIs are hosted on Render for handling AI processing, PDF extraction, and API communication.

12. Future Enhancements

The following improvements can be added in future versions:

- Real page citation support
- Semantic vector search
- User authentication
- Download summary as PDF
- Dark mode support
- Cloud deployment
- Database integration
- Multiple document support

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13. Advantages of the System

- Reduces time required to read lengthy documents
- Improves document understanding using AI
- Provides quick AI-generated summaries
- Enables intelligent question answering from PDFs
- User-friendly and responsive interface
- Supports efficient academic and professional document analysis

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14. Business Applications

- Research paper analysis
- Academic document summarization
- AI-powered study assistant
- Business report summarization
- Legal and technical document review
- Knowledge management systems

15. Learning Outcomes

Through this project, the following concepts were learned and implemented:

- AI API integration
- NLP-based summarization
- Full-stack web development
- REST API handling
- PDF parsing and processing

- React component architecture
- State management
- Frontend-backend integration
- PDF export generation

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16. Conclusion

The AI-Based PDF & Text Summarization Platform successfully demonstrates the integration of Artificial Intelligence with full-stack web development.

The project provides an efficient solution for document summarization and document-based question answering.

It showcases practical implementation of:

- NLP concepts
- AI model integration
- PDF text extraction
- REST API development
- React frontend development
- Full-stack application architecture

The project can be further extended into a more advanced AI document assistant platform.

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17. References

- React.js Documentation
- Node.js Documentation
- Express.js Documentation
- Groq API Documentation
- react-pdf Documentation
- pdf-parse Documentation
- MDN Web Docs

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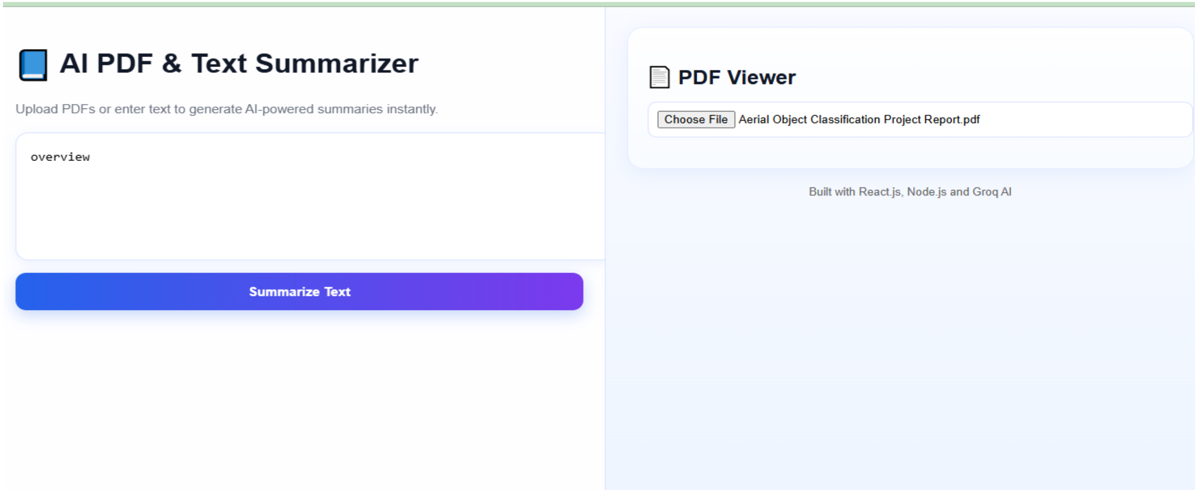
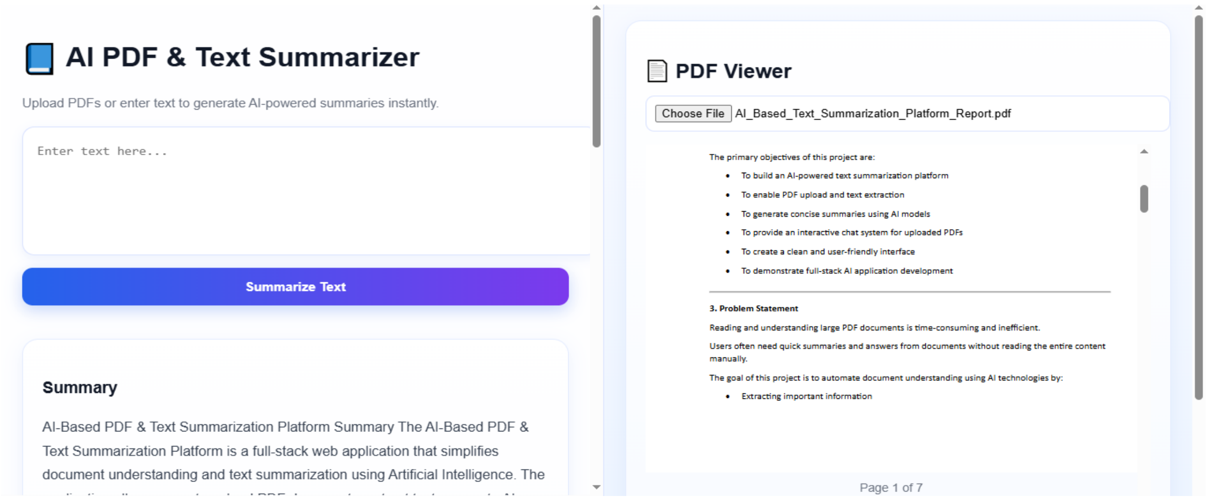
18. Screenshots

18.1 Main UI:



18.2 PDF Uploaded & Summarized:



18.3 Chat With PDF: